THE DEGREE-28 POLYNOMIAL OF A GENUS-4 LINEARISED OPERATOR, THE QUADRATIC-FORM TORSOR, AND THE SYMPLECTIC MONODROMY $\mathrm{Sp}_6(\mathbb{F}_2)$

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ABSTRACT. Let $q = 2^k$ and let $\mathfrak{q}_\lambda(x) = \mathsf{T}(x) + 2\,\mathrm{Tr}(\lambda_9 x^9 + \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x)$ be the $\mathbb{Z}/4$-valued quadratic form attached to the supersingular genus-4 curves $y^2+y = \lambda_9 x^9 + \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$ over $\mathbb{F}_q$, T the Witt-vector trace of the Teichmüller representative. Its polar operator T'_λ is a linearised operator with six-dimensional kernel $\mathcal{K}_\lambda$ and the Gauss sums of $\mathfrak{q}_\lambda$ over $\mathbb{F}_{q^{2m}}$ are governed by the coset $\mathcal{C}_\lambda = T'^{-1}_\lambda(1)$ of 64 points. We prove that $L'_\lambda = T'^8_\lambda$ satisfies $L'_\lambda(x) = x^8 B_\lambda(x)^2 + B_\lambda(x) + 1$ with the explicit polynomial $B_\lambda(x) = \lambda_9^4 x^{28} + \lambda_5^4 x^{12} + \lambda_9^2 x^{10} + \lambda_3^4 x^4 + \lambda_5^2 x^2 + \lambda_9 x + 1$, the exact analogue of the sextic of the genus-2 hexad; that

$$s_{2m}(\lambda) = \sum_{x \in \mathbb{F}_{q^{2m}}} i^{\mathfrak{q}_\lambda(x)} = q^m \sum_{w \in \mathcal{C}_\lambda \cap \mathbb{F}_{q^m}} (-1)^{[B_\lambda(w)\neq 0]}\, i^{\mathfrak{q}^{(m)}_\lambda(w)};$$

and that $\mathcal{C}_\lambda$ is canonically the torsor of quadratic forms on $\mathcal{K}_\lambda$ polarising a canonical symplectic form ω_λ, the 28 roots of B_λ being the forms of Arf invariant 1 and the 36 roots of $x^8 B_\lambda + 1$ those of Arf invariant 0. Consequently $|s_{2m}| = 8q^m$ whenever $\mathcal{C}_\lambda \subset \mathbb{F}_{q^m}$. The Frobenius acts on $\mathcal{K}_\lambda$ through $\mathrm{Sp}_6(\mathbb{F}_2)$; by a right-gcd computation in the twisted polynomial ring $\mathbb{F}_q\{\tau\}$ we determine its $\mathrm{GL}_6(\mathbb{F}_2)$-type for all λ with $q \le 512$: all 24 types of $\mathrm{Sp}_6(\mathbb{F}_2)$ occur, with the frequencies predicted by equidistribution, and the counts are not polynomial in q. An octad structure (the $S_8 = O_6^+(2)$ picture) exists only for the 79% of parameters whose Frobenius type occurs in S_8, and is never canonical; B_λ and the torsor exist for every parameter.

1. INTRODUCTION

In Parts I–IV of the author's work on prime polynomials in Legendre intervals over $\mathbb{F}_{2^k}[t]$ [4, 5, 6, 7] the degree-6 variance was reduced to the $\mathbb{Z}/4$-valued quadratic forms

$$\mathfrak{q}_\lambda(x) = \mathsf{T}(x) + 2\,\mathrm{Tr}(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x)$$

on the fields $\mathbb{F}_{q^{2m}}$, where $\mathsf{T}(x) = \mathrm{Tr}_{W_2(\mathbb{F}_{2^n})/W_2(\mathbb{F}_2)}[x]$ is the Witt-vector trace of the Teichmüller representative. These are the forms whose Gauss sums are the traces of Frobenius on the rank-four Witt–Artin–Schreier sheaves of Brylinski [3], the sheaves with which Katz [12, 13] answered the questions of Keating and Rudnick [15] on primes in short intervals of $\mathbb{F}_q[t]$ in odd characteristic; their signs are the Brown invariants of [2]. Part III showed that the radical of the polar form of $\mathfrak{q}_\lambda$ is the four-dimensional space $\mathcal{K}_\lambda$ of pairwise sums of the six roots of the sextic $B_\lambda(u) = \lambda_5^2 u^6 + \lambda_3^2 u^2 + \lambda_5 u + 1$, through the identity $L'_\lambda = x^4 B_\lambda^2 + B_\lambda + 1$ for the linearised polynomial L'_λ of the polar operator, so that the Frobenius acts on $\mathcal{K}_\lambda$ through $S_6 \cong \mathrm{Sp}_4(\mathbb{F}_2)$ (the hexad); Part IV proved the *hexad formula* expressing the Gauss sum over $\mathbb{F}_{q^{2m}}$ as a signed sum of at most sixteen fourth roots of unity indexed by the coset $T'^{-1}_\lambda(1)$, with sign $+1$ at the six roots and -1 at the ten synthematic totals.

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Code and data: https://github.com/Ruqing1963/genus4-b28-torsor.

The present paper treats the next member of the Gold family, the supersingular genus-4 curves

$$y^2+y=\lambda_9x^9+\lambda_5x^5+\lambda_3x^3+\lambda_1x,\qquad \lambda_9\neq 0, \tag{1}$$

of van der Geer and van der Vlugt [10], whose quadratic form $\mathfrak{q}_\lambda$ has a polar operator with six-dimensional kernel. The question we started from was whether the hexad of genus 2 becomes an *octad* in genus 4: the 63 nonzero vectors of $\mathbb{F}_2^6$ are the 28 duads and 35 tetrads of an 8-set, giving $S_8\cong O_6^+(2)\subset \mathrm{Sp}_6(\mathbb{F}_2)$. The answer is that the octad is the wrong structure and that the right one is simpler and more uniform. Our results are the following.

(1) The polynomial $B_\lambda(x)=\lambda_9^4x^{28}+\lambda_5^4x^{12}+\lambda_9^2x^{10}+\lambda_3^4x^4+\lambda_5^2x^2+\lambda_9x+1$ satisfies $L'_\lambda=x^8B_\lambda^2+B_\lambda+1$ (Theorem 3.1), so the 64-point coset $\mathcal{C}_\lambda=T'^{-1}_\lambda(1)$ splits canonically into the 28 roots of B_λ and the 36 roots of $x^8B_\lambda+1$.

(2) The coset-sum theorem (Theorem 4.1): $s_{2m}(\lambda)=q^m\sum_{w\in\mathcal{C}_\lambda\cap\mathbb{F}_{q^m}}\varepsilon(w)i^{\mathfrak{q}_\lambda^{(m)}(w)}$ with $\varepsilon(w)=+1$ iff $B_\lambda(w)=0$. The proof is that of Part IV with two more Newton power sums and one more squaring: the key identities are $\gamma_w^4=B_\lambda(w)$ and $(w\gamma_w)^2=\gamma_w$.

(3) The torsor theorem (Theorem 5.1): $\mathcal{K}_\lambda$ carries a canonical nondegenerate alternating form $\omega_\lambda(a,b)=\mathrm{Tr}(d_bT'_\lambda(d_a))$, and $w\mapsto Q_w$, $Q_w(k)=[\varepsilon(w+k)\neq\varepsilon(w)]$, identifies $\mathcal{C}_\lambda$ with the torsor of quadratic forms on $\mathcal{K}_\lambda$ with polar form ω_λ; $\mathrm{Arf}\,Q_w=1$ iff $B_\lambda(w)=0$. In genus 2 the same statements hold with $6=2^3-2$ and $10=2^3+2$: the hexad roots are the forms of Arf invariant 1. Hence $s_{2m}=\pm 8q^m\,i^p$ whenever $\mathcal{C}_\lambda\subset\mathbb{F}_{q^m}$ (Corollary 5.2).

(4) The Frobenius $x\mapsto x^q$ acts on $\mathcal{K}_\lambda$ as an element of $\mathrm{Sp}(\mathcal{K}_\lambda,\omega_\lambda)\cong\mathrm{Sp}_6(\mathbb{F}_2)$. Its conjugacy class in $\mathrm{GL}_6(\mathbb{F}_2)$ (its *type*) is computed for every λ with $q\le 512$ by right gcds in the twisted polynomial ring $\mathbb{F}_q\{\tau\}$, without field extensions (Section 6). The 24 types of $\mathrm{Sp}_6(\mathbb{F}_2)$ all occur, with the frequencies predicted by equidistribution in $\mathrm{Sp}_6(\mathbb{F}_2)$ to three decimals at $q=512$ (Table 1); unlike the eleven class counts of genus 2, the counts are not polynomials in q on a parity class of k.

(5) Octads (Section 7): a Frobenius-stable octad in $\mathcal{C}_\lambda$ exists only if the Frobenius type occurs in S_8, which happens for a proportion of parameters tending to $1146880/1451520=0.7901$; for $y^2+y=x^9+\lambda_1x$ over $\mathbb{F}_{2^k}$, $9\nmid k$, the Frobenius has order 9 on $\mathcal{K}_\lambda$ and no octad exists. When octads exist they are not unique (between 3 and 7 for the parameters over $\mathbb{F}_4$, $\mathbb{F}_8$ that have any), and no octic with polynomial coefficients divides $L'_\lambda+1$. The 28 roots of B_λ are the duads of every octad through a plus-type point.

Everything stated as a theorem is proved below; the statements about Frobenius types for $q\le 512$ and the frequencies are computations.

2. The form, its polar operator, and the coset

Let $q=2^k$, $\lambda=(\lambda_9,\lambda_5,\lambda_3,\lambda_1)\in\mathbb{F}_q^4$, $\lambda_9\neq 0$, $f_\lambda(x)=\lambda_9x^9+\lambda_5x^5+\lambda_3x^3+\lambda_1x$. For a finite field $E\supset\mathbb{F}_q$ of degree n over $\mathbb{F}_2$ put

$$\mathfrak{q}_\lambda^E(x)=\mathsf{T}_E(x)+2\,\mathrm{Tr}_{E/\mathbb{F}_2}(f_\lambda(x))\in\mathbb{Z}/4,\qquad \mathsf{T}_E(x)=\mathrm{Tr}_{W_2(E)/W_2(\mathbb{F}_2)}([x]),$$

and $s_E(\lambda)=\sum_{x\in E}i^{\mathfrak{q}_\lambda^E(x)}$; we write $\mathfrak{q}_\lambda^{(m)}$ and $s_m(\lambda)$ when $E=\mathbb{F}_{q^m}$. Since the first Witt component of $\mathrm{Tr}_{W_2(E)}[x]$ is $\mathrm{Tr}_{E/\mathbb{F}_2}(x)$, we have $\mathfrak{q}_\lambda^E(x)\equiv\mathrm{Tr}_{E/\mathbb{F}_2}(x)\pmod 2$. Define

$$T'_\lambda(x)=\lambda_9x^8+\lambda_5x^4+\lambda_3x^2+x+(\lambda_3x)^{1/2}+(\lambda_5x)^{1/4}+(\lambda_9x)^{1/8}, \tag{2}$$

$$L'_\lambda(x)=T'_\lambda(x)^8=\lambda_9^8x^{64}+\lambda_5^8x^{32}+\lambda_3^8x^{16}+x^8+\lambda_3^4x^4+\lambda_5^2x^2+\lambda_9x. \tag{3}$$

Lemma 2.1 (polar identity). *For $x,y\in E$: $\mathfrak{q}_\lambda^E(x+y)=\mathfrak{q}_\lambda^E(x)+\mathfrak{q}_\lambda^E(y)+2\,\mathrm{Tr}_{E/\mathbb{F}_2}\big(y\,T'_\lambda(x)\big)$.*

Proof. In W_2 of a ring of characteristic 2, $[x]+[y]=(x+y,xy)=[x+y]+V[xy]$ and $\mathrm{Tr}_{W_2(E)/W_2(\mathbb{F}_2)} V[a] = 2\,\mathrm{Tr}(a)$; so $\mathsf{T}(x+y)=\mathsf{T}(x)+\mathsf{T}(y)+2\,\mathrm{Tr}(xy)$. For a Gold monomial, $(x+y)^{2^i+1}+x^{2^i+1}+y^{2^i+1}=x^{2^i}y+xy^{2^i}$ and $\mathrm{Tr}(\lambda xy^{2^i})=\mathrm{Tr}((\lambda x)^{2^{-i}}y)$. Adding the three monomials of f_λ to $2\,\mathrm{Tr}(xy)$ gives $2\,\mathrm{Tr}(yT'_\lambda(x))$. □

Thus T'_λ is $\mathbb{F}_2$-linear, commutes with the Frobenius $x\mapsto x^q$ (its coefficients lie in $\mathbb{F}_q$), and

$$\mathcal{K}_\lambda = \ker T'_\lambda = \ker L'_\lambda, \qquad \mathcal{C}_\lambda = T'^{-1}_\lambda(1) = \{x:\ L'_\lambda(x)=1\}$$

are, over $\overline{\mathbb{F}}_2$, an $\mathbb{F}_2$-space of dimension 6 (as L'_λ is a separable 2-linearised polynomial of 2-degree 6: its derivative is $\lambda_9\neq 0$) and an affine space under it with 64 points. $\mathcal{K}_\lambda$ is the radical of the polar form of $\mathfrak{q}^E_\lambda$ on every E, and the Frobenius acts on $\mathcal{K}_\lambda$ and on $\mathcal{C}_\lambda$.

Lemma 2.2. *$\mathfrak{q}^E_\lambda$ restricted to $\mathcal{K}_\lambda\cap E$ is additive with values in $\{0,2\}$; hence $\mathrm{Tr}_{E/\mathbb{F}_2}(k)=0$ for all $k\in\mathcal{K}_\lambda\cap E$, and all points of $\mathcal{C}_\lambda\cap E$ have the same absolute trace.*

Proof. By Lemma 2.1 the polar form vanishes on $\mathcal{K}_\lambda$, so $\mathfrak{q}^E_\lambda|_{\mathcal{K}_\lambda\cap E}$ is a homomorphism from an elementary abelian 2-group to $\mathbb{Z}/4$, hence 2-torsion valued. As $\mathfrak{q}\equiv\mathrm{Tr}\pmod 2$, $\mathrm{Tr}(k)=0$. Differences of points of $\mathcal{C}_\lambda$ lie in $\mathcal{K}_\lambda$. □

3. The polynomial of degree 28

Theorem 3.1. *Let $B_\lambda(x)=\lambda_9^4x^{28}+\lambda_5^4x^{12}+\lambda_9^2x^{10}+\lambda_3^4x^4+\lambda_5^2x^2+\lambda_9x+1\in\mathbb{F}_q[x]$. Then*

$$L'_\lambda(x) = x^8B_\lambda(x)^2+B_\lambda(x)+1, \qquad i.e. \qquad L'_\lambda+1 = B_\lambda\cdot(x^8B_\lambda+1). \tag{4}$$

Proof. $x^8B_\lambda^2=\lambda_9^8x^{64}+\lambda_5^8x^{32}+\lambda_9^4x^{28}+\lambda_3^8x^{16}+\lambda_5^4x^{12}+\lambda_9^2x^{10}+x^8$; adding B_λ cancels the terms x^{28},x^{12},x^{10} and leaves $L'_\lambda+1$. □

Corollary 3.2. *The two factors in (4) are coprime and $L'_\lambda+1$ is separable, so $\mathcal{C}_\lambda$ is the disjoint union of the 28 roots of B_λ and the 36 roots of $x^8B_\lambda+1$, all simple. Both sets are stable under the Frobenius.*

Remark 3.3. $Y=x^8B_\lambda$ is a polynomial solution of $Y^2+Y=x^8(L'_\lambda+1)$; the only other solution is $Y+1$, not divisible by x^8. So B_λ is determined by L'_λ, and is produced by the coefficient recursion $b_d=c_d+b^2_{(d-8)/2}$ (c_d the coefficients of $L'_\lambda+1$), which terminates because of the shape (3). The same recursion with x^4 in place of x^8 gives the sextic of Part III from the genus-2 operator, and with x^2 the linear polynomial $B=\lambda_3x+1$ for the elliptic curves $y^2+y=\lambda_3x^3+\lambda_1x$ ($L'=\lambda_3^2x^4+x^2+\lambda_3x=x^2B^2+B+1$). For the Gold family with exponents 2^i+1, $i\le n$, one obtains $L'=x^{2^n}B^2+B+1$ with $\deg B=2^{2n-1}-2^{n-1}$, the number of quadratic forms of Arf invariant 1 on $\mathbb{F}_2^{2n}$ with a given polar form; Theorem 5.1 explains why. The three levels are linked by specialisation: setting the top coefficient to zero squares everything,

$$L'_{(0,\lambda_5,\lambda_3)} = \big(L'^{g=2}_{(\lambda_5,\lambda_3)}\big)^2, \qquad B_{(0,\lambda_5,\lambda_3)} = \big(\lambda_5^2x^6+\lambda_3^2x^2+\lambda_5x+1\big)^2,$$
$$\big(\lambda_5^2x^6+\lambda_3^2x^2+\lambda_5x+1\big)\big|_{\lambda_5=0} = (\lambda_3x+1)^2,$$

as one reads off from the formulas (Figure 1).

4. The coset-sum theorem

Theorem 4.1. *Let $m\ge 1$ and put $\varepsilon(w)=+1$ if $B_\lambda(w)=0$ and $\varepsilon(w)=-1$ otherwise. Then*

$$s_{2m}(\lambda) = q^m \sum_{w\in\mathcal{C}_\lambda\cap\mathbb{F}_{q^m}} \varepsilon(w)\, i^{\mathfrak{q}^{(m)}_\lambda(w)}. \tag{5}$$

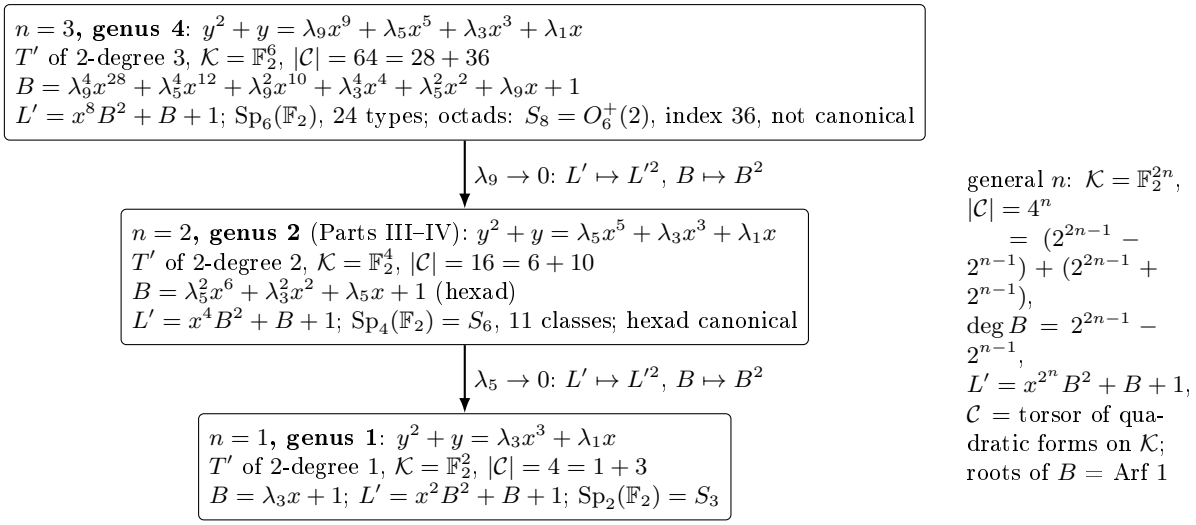


FIGURE 1. The tower of Gold families: the canonical polynomial B of the coset, the identity $L'=x^{2^n}B^2+B+1$, the $28+36$, $6+10$, $1+3$ splittings of the coset into forms of Arf invariant 1 and 0, and the specialisations that square B and L'.

Moreover, for any $c\in\mathbb{F}_{q^{2m}}$ with $c+c^{q^m}=w\in\mathcal{C}_\lambda\cap\mathbb{F}_{q^m}$ one has $\mathfrak{q}_\lambda^{(2m)}(c)=\mathfrak{q}_\lambda^{(m)}(w)+2[\varepsilon(w)=-1]$, and the sum of $i^{\mathfrak{q}_\lambda^{(2m)}}$ over a coset $c+\mathbb{F}_{q^m}$ with $c+c^{q^m}\notin\mathcal{C}_\lambda$ vanishes.

Proof. Write $V=\mathbb{F}_{q^{2m}}$, $U=\mathbb{F}_{q^m}$, $\mathfrak{q}=\mathfrak{q}_\lambda^{(2m)}$, $\mathrm{Tr}=\mathrm{Tr}_{V/\mathbb{F}_2}$, $\varphi(x)=x^{q^m}$, so $\mathrm{Tr}_{V/U}=\varphi+1$.

Step 1 ($\mathfrak{q}$ on U). For $u\in U$, $\mathsf{T}_V(u)=\mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}([u]+[u])=\mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}(0,u^2)=2\,\mathrm{Tr}_{U/\mathbb{F}_2}(u)$, and $2\,\mathrm{Tr}(f_\lambda(u))=4\,\mathrm{Tr}_{U/\mathbb{F}_2}(f_\lambda(u))=0$. So $\mathfrak{q}(u)=2\,\mathrm{Tr}_{U/\mathbb{F}_2}(u)$.

Step 2 (cosets of U). For $c\in V$, $u\in U$, Lemma 2.1 and Step 1 give $\mathfrak{q}(c+u)=\mathfrak{q}(c)+2\,\mathrm{Tr}_{U/\mathbb{F}_2}\big(u\,(1+(\varphi+1)T'_\lambda(c))\big)$, hence $\sum_{u\in U}i^{\mathfrak{q}(c+u)}=q^m\,i^{\mathfrak{q}(c)}\,[(\varphi+1)T'_\lambda(c)=1]$ and $\mathfrak{q}$ is constant on the cosets that contribute. Since T'_λ commutes with φ, $(\varphi+1)T'_\lambda(c)=T'_\lambda(c+c^{q^m})$, and $\varphi+1$ maps V onto U with kernel U; so the contributing cosets correspond bijectively to $w=c+c^{q^m}\in\mathcal{C}_\lambda\cap U$ and $s_{2m}=q^m\sum_w i^{\mathfrak{q}(c_w)}$. It remains to prove $\mathfrak{q}(c)=\mathfrak{q}_\lambda^{(m)}(w)+2[B_\lambda(w)\neq0]$.

Step 3 (the Witt value of c). Let $N=c\,c^{q^m}\in U$. By transitivity of the Witt trace and $\varphi[c]=[c^{q^m}]$, $\mathsf{T}_V(c)=\mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}([c]+[c^{q^m}])=\mathrm{Tr}_{W_2(U)/\mathbb{Z}/4}(w,N)=\mathsf{T}_U(w)+2\,\mathrm{Tr}_{U/\mathbb{F}_2}(N)$. The power sums $p_j=c^j+c^{jq^m}$ satisfy $p_j=wp_{j-1}+Np_{j-2}$, $p_0=0$, $p_1=w$, whence $p_3=w^3+Nw$, $p_5=w^5+Nw^3+N^2w$, $p_9=w^9+Nw^7+N^2w^5+N^4w$. Using $\mathrm{Tr}_{U/\mathbb{F}_2}(N^{2^j}a)=\mathrm{Tr}_{U/\mathbb{F}_2}(Na^{2^{-j}})$,

$$\mathfrak{q}(c)=\mathfrak{q}_\lambda^{(m)}(w)+2\,\mathrm{Tr}_{U/\mathbb{F}_2}(N\gamma_w),$$
$$\gamma_w=1+\lambda_9w^7+(\lambda_9w^5)^{1/2}+(\lambda_9w)^{1/4}+\lambda_5w^3+(\lambda_5w)^{1/2}+\lambda_3w.$$

Squaring twice,

$$\gamma_w^2=1+\lambda_9^2w^{14}+\lambda_9w^5+(\lambda_9w)^{1/2}+\lambda_5^2w^6+\lambda_5w+\lambda_3^2w^2,\qquad\gamma_w^4=B_\lambda(w).$$

From $T'_\lambda(w)=1$, i.e. $\lambda_9w^8+\lambda_5w^4+\lambda_3w^2+w=1+(\lambda_3w)^{1/2}+(\lambda_5w)^{1/4}+(\lambda_9w)^{1/8}$,

$$w\gamma_w=1+(\lambda_3w)^{1/2}+(\lambda_5w)^{1/4}+(\lambda_9w)^{1/8}+(\lambda_9w^7)^{1/2}+(\lambda_9w^5)^{1/4}+(\lambda_5w^3)^{1/2},$$

and $(w\gamma_w)^4=1+\lambda_3^2w^2+\lambda_5w+(\lambda_9w)^{1/2}+\lambda_9^2w^{14}+\lambda_9w^5+\lambda_5^2w^6=\gamma_w^2$. Squaring being injective, $(w\gamma_w)^2=\gamma_w$, so $\gamma_w\in\{0,w^{-2}\}$ ($w\neq0$ as $T'_\lambda(0)=0$), and $\gamma_w=0$ iff $B_\lambda(w)=0$.

If $B_\lambda(w)=0$ then $\mathfrak{q}(c)=\mathfrak{q}_\lambda^{(m)}(w)$. If $B_\lambda(w)\neq 0$ then $N\gamma_w = c\,c^{q^m}/w^2 = v\,v^{q^m}$ with $v=c/w$; as $v+v^{q^m}=1$, $N\gamma_w = v^2+v$ with $v\notin U$, so the Artin–Schreier polynomial $Y^2+Y+(v^2+v)$ has no root in U, $\mathrm{Tr}_{U/\mathbb{F}_2}(v^2+v)=1$, and $\mathfrak{q}(c)=\mathfrak{q}_\lambda^{(m)}(w)+2$. □

Remark 4.2. Nothing in the proof refers to roots of B_λ as a configuration; only the Gold shape of f_λ (bilinearity of the polar form) and the identities $\gamma_w^4 = B_\lambda(w)$, $(w\gamma_w)^2=\gamma_w$ are used. In Part IV the corresponding identities are $\gamma_w^2 = B_\lambda(w)$, $(w\gamma_w)^2=\gamma_w$. The sign $\varepsilon(w)$ does not depend on λ_1, nor on the field $\mathbb{F}_{q^m}$ over which w is considered.

5. The torsor of quadratic forms

Fix m with $\mathcal{C}_\lambda\subset\mathbb{F}_{q^m}$ (any multiple of the order of the Frobenius on $\mathcal{K}_\lambda$), $V=\mathbb{F}_{q^{2m}}$, $U=\mathbb{F}_{q^m}$. For $a\in\mathcal{K}_\lambda$ let $d_a\in V$ be any *lift*, $d_a+d_a^{q^m}=a$; for $w\in\mathcal{C}_\lambda$ let c_w be a lift of w.

Theorem 5.1. (i) $\omega_\lambda(a,b)=\mathrm{Tr}_{V/\mathbb{F}_2}\big(d_b\,T'_\lambda(d_a)\big)$ *is a well-defined nondegenerate alternating form on* $\mathcal{K}_\lambda$, *independent of* m.

(ii) *For* $w\in\mathcal{C}_\lambda$, $Q_w(k)=[\varepsilon(w+k)\neq\varepsilon(w)]$ *is a quadratic form on* $\mathcal{K}_\lambda$ *with polar form* ω_λ; *explicitly*

$$Q_w(k)=\tfrac12\big[\mathfrak{q}_\lambda^{(2m)}(d_k)-\mathfrak{q}_\lambda^{(m)}(k)\big]+\mathrm{Tr}_{V/\mathbb{F}_2}\big(d_k\,T'_\lambda(c_w)\big)\pmod 2. \tag{6}$$

(iii) $Q_{w+k}=Q_w+\omega_\lambda(k,\cdot)$, *and* $w\mapsto Q_w$ *is a bijection from* $\mathcal{C}_\lambda$ *onto the set of quadratic forms on* $\mathcal{K}_\lambda$ *with polar form* ω_λ.

(iv) $\mathrm{Arf}(Q_w)=1$ (*the Arf invariant* [1]; 28 *zeros*) *iff* $\varepsilon(w)=+1$ *iff* $B_\lambda(w)=0$; $\mathrm{Arf}(Q_w)=0$ (36 *zeros*) *iff* $x^8B_\lambda(w)+1=0$.

Proof. (ii) and well-definedness. Let $c=c_w$ and $d=d_k$. By Theorem 4.1 (applied to w with lift c and to $w+k$ with lift $c+d$) and Lemma 2.1,

$$\varepsilon(w+k)\varepsilon(w)=i^{\mathfrak{q}(c+d)-\mathfrak{q}(c)-\mathfrak{q}^{(m)}(w+k)+\mathfrak{q}^{(m)}(w)}=i^{\mathfrak{q}(d)+2\,\mathrm{Tr}(dT'(c))-\mathfrak{q}^{(m)}(k)-2\,\mathrm{Tr}_{U/\mathbb{F}_2}(kT'(w))},$$

with $T'(w)=1$ and $\mathrm{Tr}_{U/\mathbb{F}_2}(k)=0$ (Lemma 2.2). All exponents are even ($\mathfrak{q}(d)\equiv\mathrm{Tr}_{V/\mathbb{F}_2}(d)=\mathrm{Tr}_{U/\mathbb{F}_2}(k)=0$, and $\mathfrak{q}^{(m)}(k)\in\{0,2\}$), which gives (6). Replacing d by $d+u$, $u\in U$, changes $\mathfrak{q}(d)$ by $\mathfrak{q}(u)+2\,\mathrm{Tr}(uT'(d))=2\,\mathrm{Tr}_{U/\mathbb{F}_2}(u)+2\,\mathrm{Tr}_{U/\mathbb{F}_2}(u\,T'(k))=2\,\mathrm{Tr}_{U/\mathbb{F}_2}(u)$ and $\mathrm{Tr}(dT'(c))$ by $\mathrm{Tr}_{U/\mathbb{F}_2}(u\,T'(w))=\mathrm{Tr}_{U/\mathbb{F}_2}(u)$; the changes cancel. So the right side of (6) is a function of k; as a function of d it is the sum of the quadratic form $\frac12[\mathfrak{q}(d)-\mathfrak{q}^{(m)}(d+d^{q^m})]$ (note $\mathfrak{q}^{(m)}|_{\mathcal{K}_\lambda}$ is additive), whose polar form is $\mathrm{Tr}(d'T'(d))$, and of a linear form. Hence Q_w is a quadratic form on $\mathcal{K}_\lambda\cong\widetilde{V}/U$, $\widetilde{V}=\{d\in V: d+d^{q^m}\in\mathcal{K}_\lambda\}$, with polar form ω_λ, which is thereby well defined.

(i) Alternating: $2\,\mathrm{Tr}(dT'(d))=\mathfrak{q}(2d)-2\mathfrak{q}(d)\equiv 2\mathfrak{q}(d)$, so $\omega_\lambda(a,a)\equiv\mathfrak{q}(d_a)\equiv\mathrm{Tr}_{U/\mathbb{F}_2}(a)=0$. Nondegenerate: the symmetric bilinear form $\beta(x,y)=\mathrm{Tr}(yT'(x))$ on V has radical $\mathcal{K}_\lambda$ (the trace form is nondegenerate). $\widetilde{V}$ has dimension $mk+6$ and contains U and $\mathcal{K}_\lambda$; $U\subset\widetilde{V}^\perp$ because $\beta(u,d)=\mathrm{Tr}_{U/\mathbb{F}_2}(u\,T'(d+d^{q^m}))=0$; and $\dim\widetilde{V}^\perp=\dim V-\dim\widetilde{V}+\dim(\widetilde{V}\cap\mathcal{K}_\lambda)=2mk-(mk+6)+6=mk=\dim U$. So $\widetilde{V}^\perp=U$, $\mathrm{rad}(\beta|_{\widetilde{V}})=U$, and β descends to the nondegenerate form ω_λ on $\widetilde{V}/U\cong\mathcal{K}_\lambda$. Independence of m: ε is independent of m by Theorem 4.1, hence so are Q_w and its polar form.

(iii) $Q_{w+k}(k')=[\varepsilon(w+k+k')\neq\varepsilon(w+k)]=Q_w(k+k')+Q_w(k)=Q_w(k')+\omega_\lambda(k,k')$. The quadratic forms with polar form ω_λ form a torsor under $\mathcal{K}_\lambda^*\cong\mathcal{K}_\lambda$ (via ω_λ), and $w\mapsto Q_w$ is an equivariant map of $\mathcal{K}_\lambda$-torsors, hence bijective.

(iv) $\sum_{k\in\mathcal{K}_\lambda}\varepsilon(w+k)=\varepsilon(w)\sum_k(-1)^{Q_w(k)}=\varepsilon(w)(-1)^{\mathrm{Arf}\,Q_w}2^3$, the Gauss sum of a nondegenerate quadratic form on $\mathbb{F}_2^6$. The left side is $\sum_{w'\in\mathcal{C}_\lambda}\varepsilon(w')=28-36=-8$ by Corollary 3.2. Hence $\varepsilon(w)(-1)^{\mathrm{Arf}\,Q_w}=-1$. □

Corollary 5.2 (sixty-four-point lemma). *If $\mathcal{C}_\lambda \subset \mathbb{F}_{q^m}$ then, for every $\lambda_1 \in \mathbb{F}_q$, $s_{2m}(\lambda) = \pm 8\, q^m\, i^p$, where $p = \mathfrak{q}_\lambda^{(m)}(w_0)$ for any $w_0 \in \mathcal{C}_\lambda$ ($\mathfrak{q}_\lambda^{(m)}$ is constant modulo 2 on $\mathcal{C}_\lambda$).*

Proof. By Lemmas 2.1 and 2.2, $\mathfrak{q}_\lambda^{(m)}(w_0 + k) = \mathfrak{q}_\lambda^{(m)}(w_0) + 2\ell(k)$ with $\ell = \frac{1}{2}\mathfrak{q}_\lambda^{(m)}|_{\mathcal{K}_\lambda}$ linear. Then $\sum_{\mathcal{C}_\lambda} \varepsilon(w) i^{\mathfrak{q}^{(m)}(w)} = \varepsilon(w_0) i^{\mathfrak{q}^{(m)}(w_0)} \sum_k (-1)^{(Q_{w_0}+\ell)(k)} = \pm 8\, i^{\mathfrak{q}^{(m)}(w_0)}$. □

For $q = 2$ one finds $s_{14}(1,1,1,\lambda_1) = \mp 1024\, i$ and $s_{18}(1,0,0,\lambda_1) = \pm 4096\, i$ ($\lambda_1 = 0, 1$), in accordance with $8 \cdot 2^7$ and $8 \cdot 2^9$. Figure 2 shows the torsor for $\lambda = (1,1,1)$ over $\mathbb{F}_2$ in symplectic coordinates adapted to a plus-type point, with the 28 roots of B_λ, a transvection, and the action of the Frobenius.

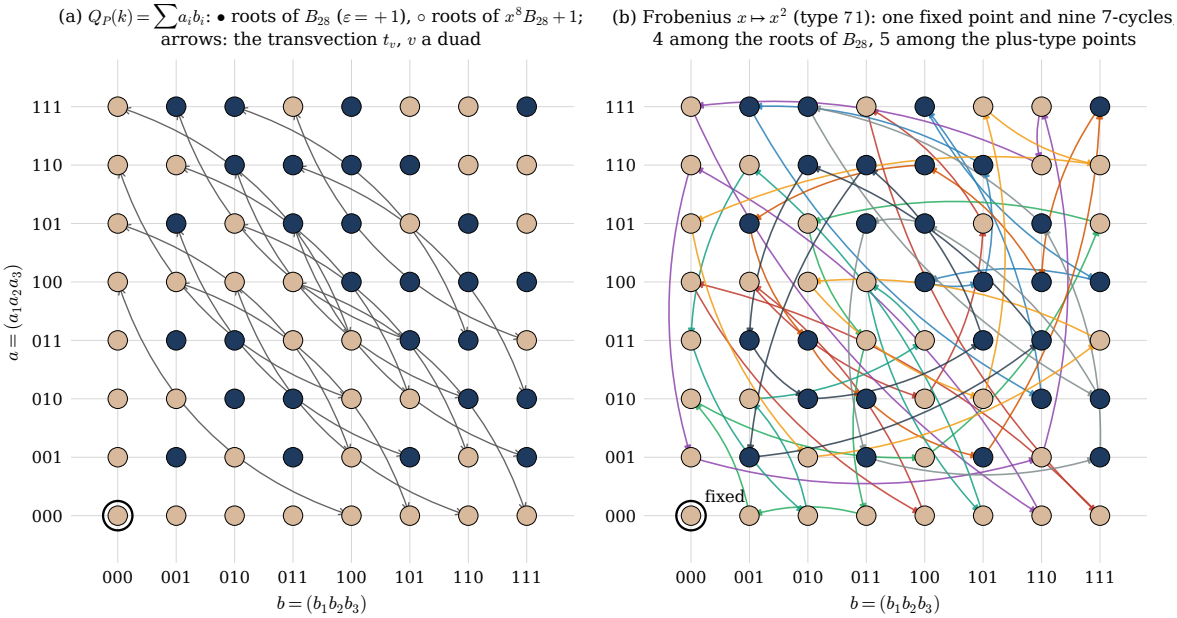


FIGURE 2. The coset $\mathcal{C}_\lambda$ for $\lambda = (1,1,1)$ over $\mathbb{F}_2$ ($\mathcal{C}_\lambda$ = the trace-one elements of $\mathbb{F}_{128}$, $\mathcal{K}_\lambda$ = the trace-zero hyperplane). A plus-type point P (ringed) and a symplectic basis $e_1, f_1, e_2, f_2, e_3, f_3$ of $(\mathcal{K}_\lambda, \omega_\lambda)$ with $Q_P(e_i) = Q_P(f_i) = 0$ are chosen; the point $P + \sum(a_i e_i + b_i f_i)$ is drawn at row a, column b, so that $Q_P = \sum a_i b_i$. Dark discs: the 28 roots of B_λ ($\varepsilon = +1$, Arf invariant 1); light discs: the 36 roots of $x^8 B_\lambda + 1$ ($\varepsilon = -1$). (a) The transvection t_v at a duad v (a difference of two roots of B_λ) exchanges the points joined by arrows and preserves both sets. (b) The Frobenius $x \mapsto x^2$, of type 71, has one fixed point and nine 7-cycles, four inside the roots of B_λ and five inside the plus-type points.

Theorem 5.3 (trace parity). *For every $m \geq 1$ and every $w \in \mathcal{C}_\lambda \cap \mathbb{F}_{q^m}$,*

$$\mathrm{Tr}_{q^m/2}(w) \equiv mk + f_m \pmod 2, \qquad f_m = \dim_{\mathbb{F}_2}(\mathcal{K}_\lambda \cap \mathbb{F}_{q^m}).$$

Proof. Put $U = \mathbb{F}_{q^m}$, $N = mk$, $K_m = \mathcal{K}_\lambda \cap U$. For $\mu \in U$ let $\mathfrak{q}'(x) = \mathfrak{q}_\lambda^{(m)}(x) + 2\,\mathrm{Tr}_{U/\mathbb{F}_2}(\mu x)$; this is a $\mathbb{Z}/4$-valued quadratic form on U with the same polar form $2\,\mathrm{Tr}(yT'_\lambda(x))$, hence with radical K_m, on which it is additive with values in $\{0, 2\}$ (Lemma 2.2). Choose μ with $\mathrm{Tr}_{U/\mathbb{F}_2}(\mu k) = \frac{1}{2}\mathfrak{q}_\lambda^{(m)}(k)$ for $k \in K_m$ (possible, since $k \mapsto \mathrm{Tr}_{U/\mathbb{F}_2}(\mu k)$ runs through all linear forms of U); then $\mathfrak{q}'$ vanishes on K_m, and by Brown's theorem [2] $s' = \sum_{x \in U} i^{\mathfrak{q}'(x)}$ satisfies $|s'|^2 = 2^{N + f_m}$. For $w \in \mathcal{C}_\lambda \cap U$, Lemma 2.1 and the symmetry of the polar form give $\mathfrak{q}'(x + w) = \mathfrak{q}'(x) + \mathfrak{q}'(w) + 2\,\mathrm{Tr}_{U/\mathbb{F}_2}(xT'_\lambda(w)) = \mathfrak{q}'(x) + \mathfrak{q}'(w) + 2\,\mathrm{Tr}_{U/\mathbb{F}_2}(x)$, and $i^{2\,\mathrm{Tr}(x)} = (-1)^{\mathfrak{q}'(x)} = i^{-2\mathfrak{q}'(x)}$; summing over x, $s' = i^{\mathfrak{q}'(w)}\,\overline{s'}$, so $s'^2 = i^{\mathfrak{q}'(w)} 2^{N + f_m}$. Write

$s' = a + bi$ with $a, b \in \mathbb{Z}$. If $\mathfrak{q}'(w)$ is even, $ab = 0$ and $a^2 - b^2 = \pm 2^{N+f_m}$ forces $N + f_m$ even; if $\mathfrak{q}'(w)$ is odd, $a^2 = b^2$ and $2ab = \pm 2^{N+f_m}$ forces $N+f_m$ odd. Thus $\mathfrak{q}'(w) \equiv N+f_m$, and $\mathfrak{q}'(w) \equiv \mathrm{Tr}_{U/\mathbb{F}_2}(w) \pmod 2$. □

Remark 5.4. In Part IV, Theorem 3.5, the same statement for genus 2 was proved under the hypothesis that $s_m(\lambda_1') \neq 0$ for some $\lambda_1' \in \mathbb{F}_q$; taking the auxiliary linear term μ in $\mathbb{F}_{q^m}$ instead of $\mathbb{F}_q$ removes the hypothesis there as well. The value $mk + f_m$ depends only on the Frobenius type: $f_m = \dim \ker(F^m - 1)$ on $\mathcal{K}_\lambda$. The congruence was checked on every rational point of every coset for $q = 2, 4, 8$ and all $m \leq n$ (script `trace_parity_g4.py`).

Remark 5.5 (genus 2)**.** With the quartic operator of Part III the same proof gives: ω_λ canonical on the four-dimensional $\mathcal{K}_\lambda$, $\mathcal{C}_\lambda$ the torsor of quadratic forms, the six roots of the sextic B_λ the forms of Arf invariant 1 ($6 = 2^3 - 2$ zeros), the ten synthematic totals the forms of Arf invariant 0 ($10 = 2^3 + 2$). The hexad formula of Part IV is thus the statement $\varepsilon(w) = -(-1)^{\mathrm{Arf}\, Q_w}$, and the sixteen-point lemma of Part IV is Corollary 5.2 with ± 4.

6. The Frobenius on $\mathcal{K}_\lambda$

By Theorem 5.1 the Frobenius $F : x \mapsto x^q$ acts on $\mathcal{K}_\lambda$ as an element of $\mathrm{Sp}(\mathcal{K}_\lambda, \omega_\lambda) \cong \mathrm{Sp}_6(\mathbb{F}_2)$. Its *type* is its conjugacy class in $\mathrm{GL}_6(\mathbb{F}_2)$, i.e. the elementary divisors of the $\mathbb{F}_2[z]$-module $(\mathcal{K}_\lambda, F)$; $\mathrm{Sp}_6(\mathbb{F}_2)$ has 30 conjugacy classes [8] and 24 types.

Proposition 6.1. *Let $\mathbb{F}_q\{\tau\}$ be the twisted polynomial ring, $\tau c = c^2 \tau$, and $P_\lambda = \lambda_9^8 \tau^6 + \lambda_5^8 \tau^5 + \lambda_3^8 \tau^4 + \tau^3 + \lambda_3^4 \tau^2 + \lambda_5^2 \tau + \lambda_9$ (so that $P_\lambda(x) = L'_\lambda(x)$). For $g \in \mathbb{F}_2[z]$,*

$$\dim_{\mathbb{F}_2} \ker\big(g(F)|_{\mathcal{K}_\lambda}\big) = \deg_\tau \mathrm{rgcd}\big(P_\lambda,\ g(\tau^k)\big).$$

Proof. For separable additive polynomials A, B over $\mathbb{F}_q$, $\ker A \subseteq \ker B$ iff $B = RA$ in $\mathbb{F}_q\{\tau\}$ (right divisibility), and the right ideal $A\mathbb{F}_q\{\tau\} + B\mathbb{F}_q\{\tau\}$ is principal, generated by the right gcd D, with $\ker D = \ker A \cap \ker B$ [17, 16]. Since $F = \tau^k$ and $g \in \mathbb{F}_2[z]$ commutes with τ, $g(F)$ is the additive polynomial $g(\tau^k)$, and $\ker g(F)|_{\mathcal{K}_\lambda} = \ker P_\lambda \cap \ker g(\tau^k)$. □

The ranks of $p(F)^j$ for the irreducible $p \in \mathbb{F}_2[z]$ of degree ≤ 6 and $j \leq 6$ determine the type; they are computed by right Euclid in $\mathbb{F}_q\{\tau\}$ with no field extension (script `ore_algebra_k6.py`, checked against companion matrices). The types of all elements of $\mathrm{Sp}_6(\mathbb{F}_2)$, built as the group generated by S_8 and one transvection on the 63-point space, were computed separately (`sp6_module_types.py`): there are 24, of which 20 occur in S_8 and 22 in $O_6^-(2)$; every element of $\mathrm{Sp}_6(\mathbb{F}_2)$ preserves at least one quadratic form with polar form ω.

Proposition 6.2 (computation)**.** *For every $(\lambda_5, \lambda_3) \in \mathbb{F}_q^2$, $\lambda_9 = 1$, $q = 2, 4, \ldots, 512$ (349 524 parameters), the type of F on $\mathcal{K}_\lambda$ is one of the 24 types of $\mathrm{Sp}_6(\mathbb{F}_2)$; at $q = 512$ all 24 occur, with the frequencies of Table 1. The proportion of parameters whose type occurs in S_8 is $0.500, 0.750, 0.750, 0.781, 0.773, 0.791, 0.794, 0.787, 0.790$ for $q = 2, \ldots, 512$, against $1146880/1451520 = 0.7901$ for $\mathrm{Sp}_6(\mathbb{F}_2)$. For no type and no parity of k is the count a polynomial of degree ≤ 2 in q (the quadratic through the three largest q of a parity class misses the smaller ones); the deviations from $|C|q^2/|\mathrm{Sp}_6(\mathbb{F}_2)|$ are of size $O(q)$ (e.g. 37885 against 37449.1 for the type Φ_7 at $q = 512$).*

The agreement with equidistribution in $\mathrm{Sp}_6(\mathbb{F}_2)$ (and the absence of any non-symplectic type, which Theorem 5.1 now explains) is what one expects if the geometric monodromy of the local system $\lambda \mapsto \mathcal{K}_\lambda$ is all of $\mathrm{Sp}_6(\mathbb{F}_2)$ [11, 14]; in genus 2 the eleven counts are exact polynomials (Part III, Table 2), and the failure of polynomiality here is the first qualitative difference between the two cases. Whether a finer invariant than the GL_6-type

TABLE 1. The 24 types of $\mathrm{Sp}_6(\mathbb{F}_2)$ ($\Phi_3 = z^2+z+1$, $\Phi_5 = z^4+z^3+z^2+z+1$, $\Phi_7 = (z^3+z+1)(z^3+z^2+1)$, $\Phi_9 = z^6+z^3+1$), the S_8 classes with that type, the proportion of elements of $\mathrm{Sp}_6(\mathbb{F}_2)$ of that type, and the observed proportion of parameters $(\lambda_5, \lambda_3) \in \mathbb{F}_{512}^2$ ($\lambda_9 = 1$) whose Frobenius has that type.

elementary divisors	S_8 class	$\lvert C\rvert/\lvert \mathrm{Sp}_6 \rvert$	observed, $q = 512$	octad
Φ_7	7 1	$1/7 = 0.1429$	0.1445	yes
$(z+1)^6$	8	$1/8 = 0.1250$	0.1230	yes
Φ_9	–	$1/9 = 0.1111$	0.1109	no
$(z+1)^2\Phi_5$	5 2 1	$1/10 = 0.1000$	0.1004	yes
$(z+1)^4\Phi_3$	4 3 1	$1/12 = 0.0833$	0.0839	yes
$(z+1)^2\Phi_3^2$	6 1 1	$1/12 = 0.0833$	0.0838	yes
Φ_3^3 (cyclic)	–	$1/12 = 0.0833$	0.0835	no
$\Phi_3\Phi_5$	5 3	$1/15 = 0.0667$	0.0660	yes
$(z+1), (z+1), \Phi_5$	$5\,1^3$	$1/30 = 0.0333$	0.0336	yes
$(z+1)^2, (z+1)^4$	4 2 1 1	$1/32 = 0.0313$	0.0309	yes
$(z+1)^2, (z+1)^2, \Phi_3$	3 2 2 1	$1/36 = 0.0278$	0.0279	yes
$(z+1), (z+1), \Phi_3^2$	6 2	$1/36 = 0.0278$	0.0281	yes
$(z+1)^2, \Phi_3, \Phi_3$	3 3 2	$1/36 = 0.0278$	0.0274	yes
Φ_3, Φ_3^2	–	$1/72 = 0.0139$	0.0142	no
$(z+1), (z+1), (z+1)^4$	4 2 2, $4\,1^4$	$1/96 = 0.0104$	0.0105	yes
$(z+1)^3, (z+1)^3$	4 4	$1/96 = 0.0104$	0.0102	yes
$(z+1), (z+1), \Phi_3, \Phi_3$	3 3 1 1	$1/108 = 0.0093$	0.0092	yes
$(z+1), (z+1), (z+1)^2, \Phi_3$	$3\,2\,1^3$	$1/144 = 0.0069$	0.0065	yes
$(z+1)^2, (z+1)^2, (z+1)^2$	$2^3\,1^2$	$1/384 = 0.0026$	0.0027	yes
Φ_3, Φ_3, Φ_3	–	$1/648 = 0.0015$	0.0015	no
$(z+1), (z+1), (z+1)^2, (z+1)^2$	2^4, $2^2\,1^4$	$1/1152 = 0.0009$	0.0008	yes
$(z+1)^{\times 4}, \Phi_3$	$3\,1^5$	$1/2160 = 0.0005$	0.0004	yes
$(z+1)^{\times 4}, (z+1)^2$	$2\,1^6$	$63/\lvert \mathrm{Sp}_6 \rvert = 4.3 \cdot 10^{-5}$	$9/2^{18} = 3.4 \cdot 10^{-5}$	yes
$(z+1)^{\times 6}$	1^8	$1/\lvert \mathrm{Sp}_6 \rvert = 6.9 \cdot 10^{-7}$	$1/2^{18} = 3.8 \cdot 10^{-6}$	yes

(the Sp_6-class, or the Frobenius orbit structure on the $28 + 36$ points) has polynomial counts is open.

7. OCTADS

Let E be an 8-set. The even subsets of E modulo complement form $V_8 = \mathbb{F}_2^6$; its 63 nonzero vectors are the 28 duads and 35 tetrads; $\langle x, y\rangle = |x \cap y| \bmod 2$ is a symplectic form and $Q_8(x) = |x|/2 \bmod 2$ a quadratic form of Arf invariant 0 (singular points the tetrads), and $S_8 = O(Q_8) = O_6^+(2)$ has index 36 in $\mathrm{Sp}(V_8)$; the odd subsets modulo complement form the affine space of 8 singletons and 56 triads. (For a 6-set, $Q_6 = |x|/2$ is not well defined and $S_6 = \mathrm{Sp}_4(\mathbb{F}_2)$: genus 2 never sees an orthogonal group.) An *octad* in $\mathcal{C}_\lambda$ is a set of eight points $e_1, \dots, e_8$ with $\sum e_i = 0$ whose duads and tetrads exhaust $\mathcal{K}_\lambda \setminus 0$; it is the same as an identification $\mathcal{K}_\lambda \cong V_8$ together with a base point.

Proposition 7.1. (a) *A Frobenius-stable octad exists in $\mathcal{C}_\lambda$ only if the type of F occurs in S_8. For $\lambda = (1,0,0)$ over $\mathbb{F}_{2^k}$ with $9 \nmid k$, F has order 9 on $\mathcal{K}_\lambda$ and no element of S_8 has order 9: no octad exists. The polynomial $L'_{(1,0,0)} + 1 = x^{64} + x^8 + x + 1$ factors over $\mathbb{F}_2$ into one linear and seven irreducible nonic factors, so no octic with coefficients in $\mathbb{F}_2[\lambda_5, \lambda_3]$ divides $L'_\lambda + 1$.*

(b) *For the parameters over $\mathbb{F}_4$ and $\mathbb{F}_8$ whose type occurs in S_8, the number of Frobenius-stable octads is* 7 *(type* 7 1*),* 6 *(*5 2 1*),* 4 *(*4 3 1*),* 3 *(*5 3*) or* 0 *(types* 8, 6 1 1, 6 2, 4 4, 4 2 1 1, $5\,1^3$*); their octics have no common shape.*

(c) *If $P \in \mathcal{C}_\lambda$ is a plus-type point, then for every octad through P (identification $\mathcal{K}_\lambda \cong V_8$ with $Q_P = Q_8$) the set $P + \{duads\}$ is the set of roots of B_λ and $P + \{0\} \cup \{tetrads\}$ the set of roots of $x^8 B_\lambda + 1$.*

Proof. (a) F permutes a stable octad, so its action on $\mathcal{K}_\lambda \cong V_8$ is that of an element of S_8. For $\lambda = (1,0,0)$, $\mathcal{K}_\lambda \cong \mathbb{F}_2[z]/(\Phi_9)$ as an $\mathbb{F}_2[F]$-module (Φ_9 is the 2-associate of $x^{64}+x^8+x$), so F has order 9 for $9 \nmid k$; the factorisation is a computation. A polynomial family of octics would specialise to a degree-8 factor at $(0,0)$. (b) is a computation (all Frobenius-stable octads are enumerated from the Frobenius orbits on $\mathcal{C}_\lambda$). (c) By Theorem 5.1(iii), $\varepsilon(P+k) = -\varepsilon(P)$ iff $Q_P(k) = 1$ iff k is a duad; $\varepsilon(P) = -1$. □

So the 28 roots of B_λ are the duads of any octad through a plus-type point, which is why they, and not eight points, form the canonical configuration; and a parameter has octads only when its Frobenius type lies in S_8, which Proposition 6.2 says happens for 79% of the parameters, while B_λ and the torsor exist for all of them.

8. The sign laws on the 24 Frobenius types

Let n be the order of F on $\mathcal{K}_\lambda$ and, for $m \geq 1$, $K_m = \mathcal{K}_\lambda \cap \mathbb{F}_{q^m} = \ker(F^m - 1)$, $f_m = \dim K_m$, $\mathcal{C}_m = \mathcal{C}_\lambda \cap \mathbb{F}_{q^m}$, and $r_m = \dim \operatorname{rad}(\omega_\lambda|_{K_m})$. By Section 6 (rational points of the coset), $\mathcal{C}_m = w_r + K_m$ is never empty. For $w \in \mathcal{C}_m$ write $\ell_w(k) = \frac{1}{2}\big[\mathfrak{q}_\lambda^{(m)}(w+k) - \mathfrak{q}_\lambda^{(m)}(w)\big] = \frac{1}{2}\mathfrak{q}_\lambda^{(m)}(k)$, a linear form on K_m (Lemmas 2.1, 2.2), which depends on λ_1 through $\operatorname{Tr}_{q^m/2}(\lambda_1 k)$.

Theorem 8.1 (structure of the Gauss sums)**.** (i) *For $w_r \in \mathcal{C}_m$,*

$$s_{2m}(\lambda) = q^m\, \varepsilon(w_r)\, i^{\mathfrak{q}_\lambda^{(m)}(w_r)} \sum_{k \in K_m} (-1)^{(Q_{w_r} + \ell_{w_r})(k)}.$$

Hence $s_{2m}(\lambda) = 0$ if $Q_{w_r} + \ell_{w_r}$ is not identically zero on $\operatorname{rad}(\omega_\lambda|_{K_m})$, and otherwise $s_{2m}(\lambda) = \pm 2^{(f_m + r_m)/2} q^m\, i^{\mathfrak{q}_\lambda^{(m)}(w_r)}$.

(ii) *For fixed $(\lambda_9, \lambda_5, \lambda_3)$ and m, the set of $\lambda_1 \in \mathbb{F}_q$ with $s_{2m}(\lambda) \neq 0$ is empty or an affine subspace of $\mathbb{F}_q$ (as $\mathbb{F}_2$-space), of codimension at most r_m; and $\mathfrak{q}_\lambda^{(m)}(w_r) \bmod 2 = mk + f_m$ for all of them.*

(iii) *If $m = dt$ with $d = \gcd(m,n)$ then $\mathcal{C}_m = \mathcal{C}_d$ and $\mathfrak{q}_\lambda^{(m)} = t\,\mathfrak{q}_\lambda^{(d)}$ on $\mathcal{C}_d$; so $s_{2m}/q^m = \sum_{\mathcal{C}_d} \varepsilon(w)\, i^{t\mathfrak{q}^{(d)}(w)}$ is determined by the data for $d \mid n$, and for even t it equals $\pm\big(\#\{w \in \mathcal{C}_d : B_\lambda(w) = 0\} - \#\{w \in \mathcal{C}_d : B_\lambda(w) \neq 0\}\big)$.*

Proof. (i) is Theorems 4.1 and 5.1(ii),(iii) with $\varepsilon(w_r + k) = \varepsilon(w_r)(-1)^{Q_{w_r}(k)}$, followed by the standard evaluation of the Gauss sum of a quadratic form on a space with radical: it vanishes unless the form vanishes on the radical, and then has absolute value $2^{(\dim + \dim \operatorname{rad})/2}$. (ii) The restriction of $Q_{w_r} + \ell_{w_r}$ to $R = \operatorname{rad}(\omega_\lambda|_{K_m})$ is a linear form (its polar form vanishes on R), affine-linear in λ_1 through $\operatorname{Tr}_{q^m/2}(\lambda_1 k) = \operatorname{Tr}_{q/2}(\lambda_1 \operatorname{Tr}_{q^m/q} k)$; the condition that it vanish is a system of at most r_m affine-linear equations on λ_1. The parity statement is Theorem 5.3. (iii) The first claim holds because F^n is trivial on $\mathcal{C}_\lambda \subset \mathbb{F}_{q^n}$; the second is the multiplicity lemma of Part III ($\mathsf{T}_{\mathbb{F}_{q^{dt}}} = t\,\mathsf{T}_{\mathbb{F}_{q^d}}$ on $\mathbb{F}_{q^d}$, and the trace of f_λ likewise); for even t, $i^{t\mathfrak{q}^{(d)}(w)} = (-1)^{(t/2)\mathfrak{q}^{(d)}(w)} = (-1)^{(t/2)\operatorname{Tr}(w)}$ is independent of w on $\mathcal{C}_d$ by Theorem 5.3 (equal to $+1$ if $4 \mid t$ and to $(-1)^{\operatorname{Tr}(w)}$ if $t \equiv 2 \pmod 4$), hence pulls out of the sum. □

Remark 8.2 (the L-polynomial)**.** $s_m(\lambda)$ is minus the trace of F^m on the Witt–Artin–Schreier sheaf $\mathcal{L}_{\psi_4}(x, f_\lambda(x))$ on $\mathbb{A}^1$, which is lisse of rank $\deg f_\lambda - 1 = 8$ with Swan conductor 9 at infinity and pure of weight 1 [3, 12, 9]. Hence $L_\lambda(T) = \prod_{i=1}^{8}(1 - \omega_i T) \in$

$\mathbb{Z}[i][T]$ with $|\omega_i| = \sqrt{q}$, and $s_m = -\sum \omega_i^m$. We computed $s_1, \dots, s_{10}$ for all λ with $q = 2, 4$ (script `witt_lpoly_g4.py`): the Newton identities from $s_1, \dots, s_8$ give L_λ with Gaussian-integer coefficients, s_9, s_{10} agree with the prediction (confirming the rank), all $|\omega_i| = \sqrt{q}$, and the normalised eigenvalues $\omega_i/\sqrt{q}$ are roots of unity of order N dividing $8n$ (n the Frobenius order; e.g. $N \in \{14, 28, 56\}$ for type 7 1, $N \in \{8, 16\}$ for type 8, $N \in \{9, 36, 72\}$ for Φ_9, $N \in \{12, 24\}$ for Φ_3^3, $N \in \{30, 60, 120\}$ for 5 3). Figure 3 shows six types; the rigidity visible there (eigenvalue multisets symmetric under $\omega \mapsto i\omega$ or $\omega \mapsto -\omega$, depending on λ_1) is the spectral form of Theorem 8.1(iii): s_{2m} depends on m only through $\gcd(m, n)$ and $m/\gcd(m, n)$ mod 4.

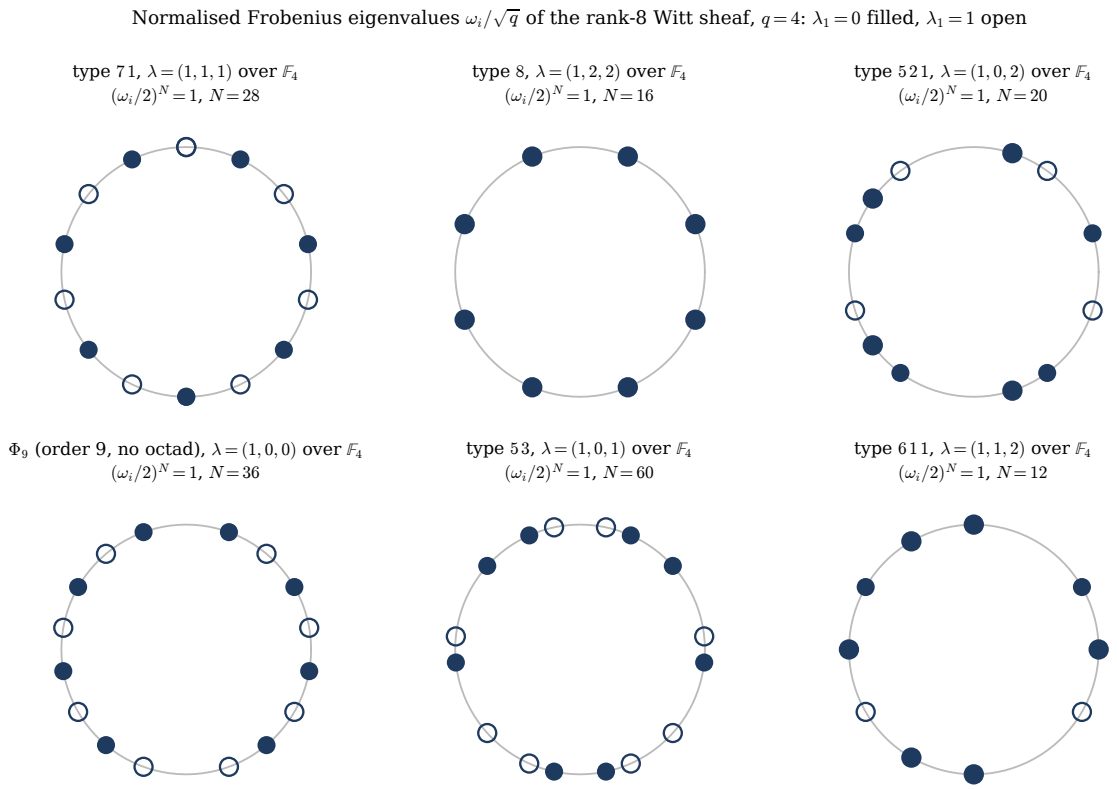


FIGURE 3. Normalised Frobenius eigenvalues $\omega_i/\sqrt{q}$ ($i = 1, \dots, 8$) of the rank-8 Witt sheaf for one parameter of each of six Frobenius types at $q = 4$, $\lambda_1 = 0$ (filled) and $\lambda_1 = 1$ (open); N is the order of the group of roots of unity they generate.

What Theorem 8.1 leaves open is the sign $\pm$ in (i), the genus-4 analogue of the sign laws of Part IV (Theorems 3.9, 3.17, 3.20 there). Table 2 records the data for all parameters with $q \le 8$ (script `genus4_sign_rules.py`): for each Frobenius type and each $d \mid n$, the invariants f_d, r_d, the number of λ_1 with $s_{2d} \neq 0$, and the set of values of s_{2d}/q^d.

9. QUESTIONS

(1) The sign in Theorem 8.1(i) as a function of λ, type by type. Table 2 separates two situations. For the full-coset sums ($d = n$) and for several smaller d the sign is the same for every parameter of the type (e.g. $W_n = -8$ for the types 8, 4 2 1 1, 4 4, 4 3 1, Φ_3^3 and $W_n = +8$ for 6 2 at $q = 8$); there the sign is $\varepsilon(w_r)(-1)^{\mathrm{Arf}(Q_{w_r}+\ell)}$ for the unique rational point w_r of $\mathcal{C}_d$ when $f_d = 0$, or an Arf invariant on K_d in general, and a

TABLE 2. Gauss sums $s_{2d}(\lambda)$ by Frobenius type, $d \mid n$, all parameters $(\lambda_5, \lambda_3) \in \mathbb{F}_q^2$ of the type and all $\lambda_1 \in \mathbb{F}_q$ ($q = 8$; a type absent at $q = 8$ is shown at the largest $q \le 8$ where it occurs). $f_d = \dim K_d$, $r_d = \dim \mathrm{rad}(\omega|_{K_d})$; "$\#\lambda_1$" is the number of $\lambda_1 \in \mathbb{F}_q$ with $s_{2d} \neq 0$ (with multiplicity over the parameters when it varies); $|s_{2d}|/q^d = 2^{(f_d+r_d)/2}$ when nonzero; $W_d = \sum_{\lambda_1 \in \mathbb{F}_q} s_{2d}/q^{d+1}$ with its multiplicity over the parameters of the type.

type	q	#par.	d	f_d	r_d	$\#\lambda_1$	$\lvert s_{2d}\rvert/q^d$	W_d
7 1	8	13	1	0	0	8	1	0
			7	6	0	8	8	0
8	8	6	1	1	1	4	2	0
			2	2	2	4	4	$+2(\times 3)$, $-2(\times 3)$
			4	4	2	8	8	0
			8	6	0	8	8	-8
5 2 1	8	9	1	1	1	4	2	0
			2	2	0	8	2	0
			5	5	1	4	8	0
			10	6	0	8	8	0
4 3 1	8	6	1	1	1	4	2	0
			2	2	2	4	4	$+2(\times 3)$, $-2(\times 3)$
			3	3	1	4	4	0
			4	4	0	8	4	$+4$
			6	4	2	4	8	$+4(\times 3)$, $-4(\times 3)$
			12	6	0	8	8	-8
6 1 1	8	3	1	1	1	4	2	0
			2	2	0	8	2	0
			3	3	3	4	8	0
			6	6	0	8	8	0
5 3	8	3	1	0	0	8	1	0
			3	2	0	8	2	0
			5	4	0	8	4	0
			15	6	0	8	8	0
5 1 1 1	8	1	1	2	0	8	2	0
			5	6	0	8	8	0
4 2 1 1	8	3	1	2	2	2	4	0
			2	4	2	4	8	0
			4	6	0	8	8	-8
6 2	8	3	1	2	0	8	2	0
			2	2	0	8	2	$+2$
			3	4	2	8	8	0
			6	6	0	8	8	$+8$
4 4	8	1	1	2	2	2	4	0
			2	4	2	8	8	-4
			4	6	0	8	8	-8
Φ_3, Φ_3, Φ_3	8	1	1	0	0	8	1	0
			3	6	0	8	8	0
Φ_9	8	9	1	0	0	8	1	0
			3	0	0	8	1	0
			9	6	0	8	8	0
Φ_3^3 (cyclic)	8	6	1	0	0	8	1	0
			2	0	0	8	1	-1
			3	2	2	8	4	0
			4	0	0	8	1	$+1$
			6	4	2	8	8	$+8$
			12	6	0	8	8	-8
Φ_3, Φ_3^2	4	1	1	0	0	4	1	0
			2	0	0	4	1	$+1$
			3	4	2	4	8	0
			6	6	0	4	8	-8

proof should follow from Theorems 5.1 and 5.3 together with the Frobenius structure of K_d. For the types 8 and 4 3 1 at $d = 2$ (and 4 3 1 at $d = 6$) the parameters split into two halves with $W_d = \pm 2$ (resp. ± 4), the shape of the order-4 classes $4\,1^2$, 4 2 of Part IV, whose sign is $(-1)^{\mathrm{Tr}(\rho_0/\tau_0^2)}$ in terms of the quartic cofactor of the sextic;

here the analogous invariant should be a quadratic character of the ground-field data of the factorisation of B_λ over $\mathbb{F}_q$ (its degree-2 and degree-4 factors for these types), and the analogue of the hyperplane parameter κ_0 is the affine condition on λ_1 in Theorem 8.1(ii). We have not found the closed form.

(2) The counts of the Frobenius types as functions of q, and the geometric monodromy of $\lambda \mapsto \mathcal{K}_\lambda$.

(3) The identification of the Legendre-interval problem that the family (1) governs (degree 10 if the pattern of Parts I–III persists; the degree-8 problem involves x^7, which has 2-weight 3, no quadratic form, and curves $y^2 + y = x^7 + \dots$ that are not supersingular: their Newton polygons over $\mathbb{F}_2$ have slopes $1/3, 2/3$).

10. Computations

All scripts are in Python (numpy, sympy, matplotlib) and are in the repository named on the first page. `octad_combinatorics.py`: the 63-point geometry, S_8, $\mathrm{Sp}_6(\mathbb{F}_2)$ by closure, orbits, element orders (Section 7). `genus_check.py`: Newton polygons of $y^2 + y = x^7 + \dots$, bilinearity tests, module types of S_8, the order-9 obstruction (Proposition 7.1(a), Section 9). `ore_algebra_k6.py`, `sp6_module_types.py`, `genus4_class_stats.py`: Proposition 6.1, Table 1, Proposition 6.2. `octad_sum_test.py`: Lemma 2.1 on $\mathbb{F}_{16}$, $\mathbb{F}_{64}$ (all λ) and the fibre sums over $\mathbb{F}_{2^{14}}$, $\mathbb{F}_{2^{18}}$ (Theorem 4.1, Corollary 5.2). `find_octic_identity.py`: Proposition 7.1(a),(b) and the rational points of the coset. `coset_signs.py`, `b28_identity.py`: ε from one lift per point, the torsor checks of Theorem 5.1 and the interpolation that first produced B_λ (all λ for $q \le 8$, samples at $q = 16, 32$, general λ_9 at $q = 4, 8$), the identities $\gamma_w^4 = B_\lambda(w)$, $(w\gamma_w)^2 = \gamma_w$ on every coset point for several parameters, and the exact division (4) in $\mathbb{F}_2[\lambda_9, \lambda_5, \lambda_3][x]$. `trace_parity_g4.py`: Theorem 5.3 on every rational point of every coset, $q \le 8$, all $m \le n$ (7880 points). `genus4_sign_rules.py`, `make_sign_table.py`: Theorem 8.1 and Table 2, the sums $s_{2d}(\lambda)$ from the coset data, the invariants f_d, r_d, the nonvanishing counts in λ_1 and the sums W_d. `witt_lpoly_g4.py`: Remark 8.2 and Figure 3, the sums $s_1, \dots, s_{10}$, the L-polynomials by Newton's identities and the normalised eigenvalues. `make_figures_b28.py`: Figures 2 and 3.

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